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Metagenomics and metabolomics to evaluate the potential role of gut microbiota and blood metabolites in patients with cerebral infarction

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Abstract
Cerebral infarction, a cerebrovascular disorder, is characterized by the sudden onset of neurological deficits and clinical symptoms. It ranks among the leading causes of death and severe disability worldwide. The etiology of cerebral infarction is multifaceted, with common risk factors including dietary patterns, smoking, hypertension, and diabetes mellitus. In recent years, the role of the gut microbiota in systemic immunity and tumorigenesis has been intensively explored, thrusting the research on the gut—brain axis into the spotlight. However, there is a lack of literature investigating the relationship between the gut microbiota and blood metabolites in cerebral infarction. In this study, we employed 16S rRNA analysis and ultra—high—performance liquid chromatography—tandem mass spectrometry (UHPLC—MS/MS) for a comprehensive metagenomic and metabolomic analysis of fecal samples from cerebral infarction patients and the general population. Our results revealed a significant correlation between the gut microbiome and serum metabolites, highlighting the impact of the microbiome on metabolic pathways. Specifically, we found that 35 gut microbiome taxa, such as Actinobacteriota and Peptostreptococcales—Tissierellales, were significantly enriched in the control group (N group). Through Linear Discriminant Analysis Effect Size (LEfSe) analysis, 72 taxa showed significant differences between cerebral infarction patients and healthy individuals. Among them, 22 key taxa were identified as microbial biomarkers for differentiating patients from healthy controls. These findings suggest that variations in the microbiome and metabolites could potentially serve as biomarkers for future diagnostic and therapeutic strategies in cerebral infarction.

Keywords Cerebral infarction, Microbiome, Metabolite, Biomarker

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Introduction

Cerebral infarction, colloquially referred to as ischemic stroke, is a medical condition precipitated by the cessation of cerebral blood flow, culminating in ischemia and hypoxic damage to brain tissue, and consequent neurological dysfunction. The clinical presentation includes abrupt onset of headache, nausea, vomiting, altered levels of consciousness, speech disturbances, weakness in the limbs, and facial asymmetry [1]. This condition is a leading cause of mortality and severe disability worldwide, exerting significant economic strain on both families and society at large. The risk factors implicated in the etiology of cerebral infarction include, but are not limited to, dietary habits, smoking, hypertension, and diabetes mellitus [2]. Effective management of these risk factors is crucial in reducing the incidence and severity of cerebral infarction.

In recent years, the gut microbiota has emerged as a focal point of extensive research, particularly within the realms of systemic immunity and oncogenesis [3, 4]. It has been well—documented that certain specific microbial species and their metabolites play a role in the pathogenesis of cancer. For example, in the development of colorectal cancer, the equilibrium of intestinal metabolites and their interaction with the microbiota is perturbed during the disease progression [5]. Additionally, comparative metagenomic and plasma metabolomic analyses on cancer patients have revealed significant differences in gut microbial composition, functional pathways, and associated plasma metabolites between those with and without cachexia, suggesting a potential role for the gut microbiota in cachexia development and its therapeutic implications [6].

The gut microbiota and the brain—gut axis have garnered substantial research interest, particularly in the context of stroke pathology. Studies have indicated that post—stroke dysbiosis within the gut microbiota can lead to heightened intestinal permeability and the subsequent activation of the intestinal immune system, which in turn can elevate systemic levels of pro—inflammatory cytokines [7]. This inflammatory milieu may facilitate the translocation of gut bacteria and pro—inflammatory cells into the brain tissue, potentially through a compromised blood—brain barrier, thereby exacerbating ischemia—reperfusion injury [8, 9].

Moreover, in the context of stroke—related research, non—targeted metabolomics methods have been used to analyze the pathogenesis of cerebral infarction. Phosphatidylcholine metabolites, such as choline, trimethylamine N—oxide (TMAO), and methionine, which are associated with cardiovascular diseases, have been investigated in this regard [10]. Also, an increase in the levels of Lactobacillus and Bifidobacterium and a decrease in butyrate—producing genes may increase the stroke risk in diabetic patients [11, 12].

The relationship between the gut microbiota and blood metabolites in patients with cerebral infarction, as well as their potential impacts on disease progression, remains an area that is not sufficiently explored within the scientific community. While some studies have delved into the relationship between the gut microbiota and stroke, the specific links between the gut microbiota and blood metabolites and their effects on disease progression still need further investigation. For instance, short—chain fatty acids (SCFAs) can be absorbed through monocarboxylate transporters (MCTs), enter the brain, reduce IL—17 + γδ T cells, decrease activated microglia, and enhance synaptic plasticity, highlighting their importance in stroke recovery [13, 14]. Although bile acid metabolism has not been directly mentioned in the current research context, it is related to the regulation of the gut microbiota and metabolites and may be a future research direction.

In this study, aiming to fill this knowledge gap, we investigated the differences in intestinal flora between patients with cerebral infarction and the general population, as well as the relationship between intestinal flora and blood metabolites. By conducting microbiome and metabolome analyses of fecal and blood samples from both groups, we observed correlations between the gut microbiome and serum metabolomics, indicating the microbiome's influence on metabolite profiles. These findings suggest that distinct microbiomes and metabolites may serve as potential biomarkers for future diagnostics and therapeutic strategies in cerebral infarction.

Method

Sample collection

We initially conducted metagenomic sequencing on 60 fecal samples, which were categorized into two groups: 30 from healthy controls (H) and 30 from patients with cerebral infarction (N). The control group consisted of individuals with no history of chronic disease. All participants were Han Chinese, born via natural delivery, had resided in the North China region, and had not taken any probiotics, prebiotics, laxatives,or antibiotics in the month prior to admission. Fresh fecal samples were immediately preserved at −80 °C to maintain their integrity.

Upon completion of the treatment phase, serum samples were obtained and processed by centrifugation at 3,000 rpm for 10 min at 4 °C. The resulting supernatant was then stored at −80 °C for subsequent analysis of biochemical markers.

DNA extraction and 16S rDNA sequencing

Fecal samples were processed to extract microbial DNA utilizing the E.Z.N.A.® Stool DNA Kit (catalog number D4015, Omega, Inc., USA), adhering to the manufacturer's guidelines. Subsequently, the V3-V4 variable region of the 16S rRNA gene from bacterial species was targeted for PCR amplification. This PCR process commenced with an initial denaturation step at 95 °C for 3 min, succeeded by 25 cycles of denaturation at 95 °C for 30 s, annealing at 55 °C for 30 s, extension at 72 °C for 30 s, and concluded with a final extension phase at 72 °C for 5 min. The amplification utilized primers 338 F (5'-ACTCCTACGGGAGGCAGCAG-3') and 806R (5'-GGACTACHVGGGTWTCTAAT-3'), each equipped with a distinct eight-base barcode for sample differentiation.

The PCR was conducted in triplicate, with each reaction consisting of a 20 μL solution that included 4 μL of 5 × FastPfu Buffer, 2 μL of 2.5 mM dNTPs, 0.8 μL of each primer at a concentration of 5 μM, 0.4 μL of FastPfu Polymerase, and 10 ng of the DNA template. The purified PCR products, known as amplicons, were then pooled to ensure equal molar ratios and sequenced using the paired-end method on an Illumina MiSeq PE300 platform (Illumina, San Diego, USA). This sequencing was executed following the standard operating procedures provided by Majorbio Bio-Pharm Technology Co., LTD. (Shanghai, China).

Processing of sequencing data

The FASTQ files were converted into a format compatible with the QIIME2 system by employing the QIIME2 tools import utility, as referenced in [15]. Subsequently, the demultiplexed sequences for each sample underwent a series of quality control measures, including filtering, trimming, and de-noising, facilitated by the QIIME2 dada2 plugin. This process culminated in the identification and elimination of chimeric sequences, resulting in a feature table of amplicon sequence variants (ASVs), as detailed in [16].

Further analysis was conducted using the QIIME2 feature-classifier plugin, which aligned the ASV sequences against a curated SILVA database, specifically tailored to the V3-V4 hypervariable region defined by the 338F/806R primer set. This alignment was instrumental in generating a comprehensive taxonomy table, as outlined in [17].

Bioinformatics analysis

The bacterial richness and diversity within the samples, collectively referred to as alpha diversity, were quantified using a suite of indices: the Chao1, ACE, and Sobs for species richness, alongside the Shannon and Invsimpson indices for species evenness and dominance. These metrics provided insights into the variety and distribution of bacterial genera and their proportional representation within the gut microbiota. Employing the phylogenetic information derived from the sequences of each sample, beta diversity was assessed using the weighted UniFrac metric and principal coordinates analysis (PCoA). This method effectively highlighted variations in the composition of gut microbial communities across different groups. The computation of alpha diversity was facilitated by the"vegan"package in R [10]. Additionally, the"encodes"R package was utilized to determine the Bray–Curtis dissimilarity, which quantifies the ecological distance between different sample types [18]. Furthermore, leveraging the Galaxy Platform (https://huttenhower.sph.harvard.edu/galaxy/) [17], we applied the linear discriminant analysis effect size (LEfSe) method to discern differentially abundant microbial groups that were indicative of specific classifications among the samples. This approach aided in identifying key bacterial taxa that were representative of the observed groupings.

Random forest model construction and validation

To determine the most influential taxa within the gut microbiome, we employed the Random Forest model, a robust ensemble learning technique suitable for both classification and regression tasks. This model was configured with the relative abundance data of bacterial taxa, utilizing the'randomForest'algorithm with a default setting of 1,000 trees (ntree = 1,000ntree = 1,000), and a metric derived from p/3p/3, where pp denotes the number of taxa in each class [19]. For the differentiation of paired groups, stepwise logistic regression models were constructed using the'glm'function from the R package'stats'. The identification of biomarkers was executed through a stepwise selection process facilitated by the'MASS'package in R [20]. Initially, all bacterial species exhibiting significant variations were considered as potential biomarkers within the models.Subsequently, the final set of biomarkers was refined using a stepwise model selection algorithm, guided by the Akaike Information Criterion (AIC). This refinement was achieved through the'STEPAIC'function from the'MASS'package. The biomarkers identified were subsequently validated using the Random Forest method with a tenfold cross-validation approach, implemented with the'caret'package in R [21]. The performance of the classification models was evaluated using the receiver operating characteristic (ROC) analysis, which was conducted with the'pROC'package in R [21]. This analysis provides a graphical representation of the models'discriminatory capabilities.

UHPLC-MS/MS analysis

Analyses via ultra-high-performance liquid chromatography-tandem mass spectrometry (UHPLC-MS/MS) were conducted using a Vanquish UHPLC setup (Thermo

Fisher Scientific, Germany) interfaced with an Orbitrap Q Exactive HF mass spectrometer (Thermo Fisher Scientific, Germany). The samples were loaded onto a Hypersil Gold analytical column (dimensions: 100 mm × 2.1 mm, particle size: 1.9 μm) and resolved using a 17-min linear solvent gradient at a flow velocity of 0.2 mL per minute.

For the positive ionization mode, the mobile phases consisted of eluent A (water containing 0.1% formic acid) and eluent B (methanol). Conversely, for the negative ionization mode, the phases were eluent A (5 mM ammonium acetate adjusted to pH 9.0) and eluent B (methanol). The gradient elution was programmed as follows: starting at 2% B for 1.5 min, increasing to 85% B over 3 min, then ramping up to 100% B over the next 10 min, followed by a return to 2% B over 10.1 min, and finally maintaining 2% B for 12 min.

The Q Exactive HF mass spectrometer was operated in both positive and negative polarity modes. Operational parameters included a spray voltage of 3.5 kilovolts, a capillary temperature set to 320 degrees Celsius, sheath gas flow at 35 pounds per square inch, auxiliary gas flow at 10 L per minute, an S-lens radio frequency level of 60, and an auxiliary gas heater temperature of 350 degrees Celsius.

Data processing and metabolite identification

The UHPLC-MS/MS-generated raw data files underwent processing through Compound Discoverer version 3.1 (Thermo Fisher Scientific), which facilitated peak alignment, identification, and quantification for each metabolite. Key parameters for this analysis were configured as follows: a retention time window of 0.2 min, a mass accuracy of 5 parts per million (ppm), a signal intensity threshold of 30%, a signal-to-noise ratio cutoff at 3, and other intensity-related criteria.

Subsequent to this, the peak intensities were standardized against the overall spectral intensity to ensure comparability across samples. This normalization step was pivotal for the subsequent prediction of molecular formulas, which was achieved by examining additive ions, molecular ion peaks, and fragment ions. The refined data were then cross-referenced with databases including mz Cloud, mz Vault, and MassList to secure precise qualitative and quantitative metabolite profiles. It is important to note that the mz Cloud database(https://www.mzcloud.org/), which was integral to our analysis.

Table 1 Comparison of demographic and clinical characteristics between N and H

Indicators	N ($n=30$)	H ($n=30$)	t value	p value
Sex(m/f)	25/5	22/8	-	0.85
Age	67.83 ± 13.22	66.00 ± 8.56	0.63	0.53
WBC	8.50 ± 1.61	7.44 ± 1.28	0.11	0.10
LYM	2.46 ± 0.69	2.94 ± 0.60	0.001	0.001
NEU	5.84 ± 1.41	4.18 ± 0.93	0.001	0.001
MONO	0.56 ± 0.14	0.54 ± 0.12	0.80	0.86
RBC	5.05 ± 0.40	4.80 ± 0.33	0.35	0.42
PLT	242.57 ± 42.27	231.13 ± 27.10	0.77	0.49
HGB	154.16 ± 11.60	150.66 ± 13.09	0.23	0.15
LDL	3.29 ± 0.76	3.66 ± 0.70	0.05	0.06
HDL	1.27 ± 0.20	1.66 ± 0.22	0.001	0.001
UA	397.10 ± 83.16	360.13 ± 67.69	0.41	0.29
GLU	7.93 ± 1.91	5.72 ± 0.48	0.06	0.73
HCY	27.49 ± 9.27	20.92 ± 8.23	0.07	0.001
TG	2.32 ± 0.89	2.30 ± 1.00	0.55	0.19
CRP	41.26 ± 25.43	2.82 ± 1.50	0.02	0.001

WBC white blood cell, *LYM* lymphocyte, *NEU* neutrocyte, *MONO* monocyte, *RBC* red blood cell, *PLT* platelet, *HGB* hemoglobin, *LDL* low density lipoprotein, *HDL* high density lipoprotein, *UA* uric acid, *GLU* glucose, *HCY* Homocysteine, *TG* triglyceride

Data analysis

The Wilcoxon rank-sum test was implemented for assessing differences in continuous variables. The statistical analyses distinguishing unique metabolites were executed utilizing R software (version 4.0.3), Python (version 2.7.6), and the CentOS operating system (release 6.6). In instances where the data exhibited non-normal distribution, it was normalized using the formula for converting raw quantitation values into relative peak areas [22]. Metabolites with coefficients of variation (CVs) in quality control (QC) samples exceeding 30% were excluded from further analysis. Following these steps, the identification and relative quantification of the metabolites were achieved. Statistical significance was considered at the threshold of $p < 0.05$.

The interactions within disease-associated microbiomes were assessed using Spearman's rank correlation [6]. Visualization of the data was performed through the generation of heatmaps, which were created using the"heatmap"package within R software [23]. Annotations for the identified metabolites were obtained from reputable databases, including the Kyoto Encyclopedia of Genes and Genomes (KEGG) (https://www.genome.jp/kegg/pathway.html) [24], the Human Metabolome Database (HMDB) (https://hmdb.ca/metabolites) [25], and the LIPID Maps database(http://www.lipidmaps.org) [26].

Results

Characteristics of all participants

Sixty individuals were recruited for this study, divided into two groups: N comprising 30 participants and H also consisting of 30 participants. The clinical characteristics of the study population are detailed in Table 1.

Alpha and beta-diversity of gut microbiota in H and N

The concept of alpha-diversity was employed to assess the variability of microbial communities within individual

groups. Our analysis of alpha-diversity indicated no significant differences in the microbial community diversity between the H group subjects and those afflicted with N, as evidenced by the Shannon, Simpson, Ace, and Chao1 indices (Fig. 1A-D).

To assess the overall structural configuration of the gut microbiota, the study constructed a principal coordinate analysis (PCoA) score plot based on unweighted UniFrac distances (Fig. 1E). This visualization approach aimed to elucidate the holistic structural and compositional characteristics of the microbiome through a phylogenetic distance matrix reflecting inter-sample differences. Samples from the two groups exhibited a clear separation trend along the PC1 axis, indicating inter-group differences in the β-diversity of the gut microbiota (Fig. 1E).

Gut microbiota disturbance relates to N progression

The amplicon sequence variants (ASVs) identified were subjected to a classification and annotation process, and a comparative analysis was performed between the intestinal samples of the patient group (N) and the healthy control group (H), as depicted in Fig. 2A. It was observed that the number of unique gut bacterial species in the patient group was substantially lower than that in the healthy group, with a count of 883 unique species in N compared to 492 in H.

To elucidate the relationship between the two groups further, a genus-level comparison of the intestinal microorganisms was conducted, as illustrated in Fig. 2B. This comparison indicated that the N patients possessed a similar count of fecal microbiotas as the H group. Although the overall community structure of the gut

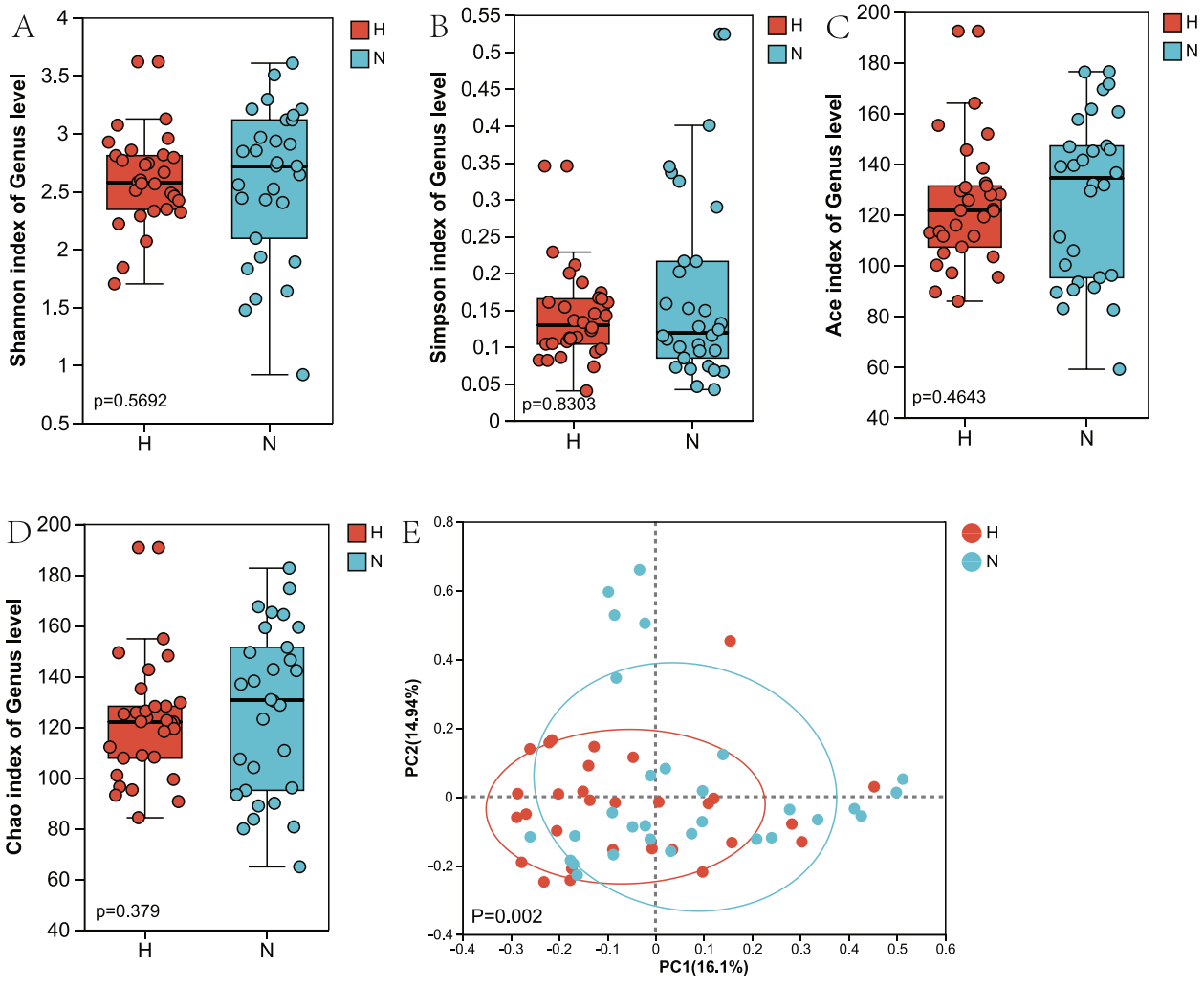

Fig. 1 Changes in bacterial diversities in gut microbiota in Cerebral infarction patients compared with healthy controls. **A** Comparison of Alpha-diversity(as assessed by the Shannon index). $P > 0.05$.Wilcoxon rank-sum test. **B** Comparison of Alpha-diversity(as assessed by the Simpson index). $P > 0.05$.Wilcoxon rank-sum test. **C** Comparison of Alpha-diversity(as assessed by the Ace index). $P > 0.05$. Wilcoxon rank-sum test. **D** Comparison of Alpha-diversity(as assessed by the Chao index). $P > 0.05$.Wilcoxon rank-sum test. **E** The plots were based on unifrac distances. Green plots represent Cerebral infarction patients(N), and blue plots represent health(H). $P < 0.05$

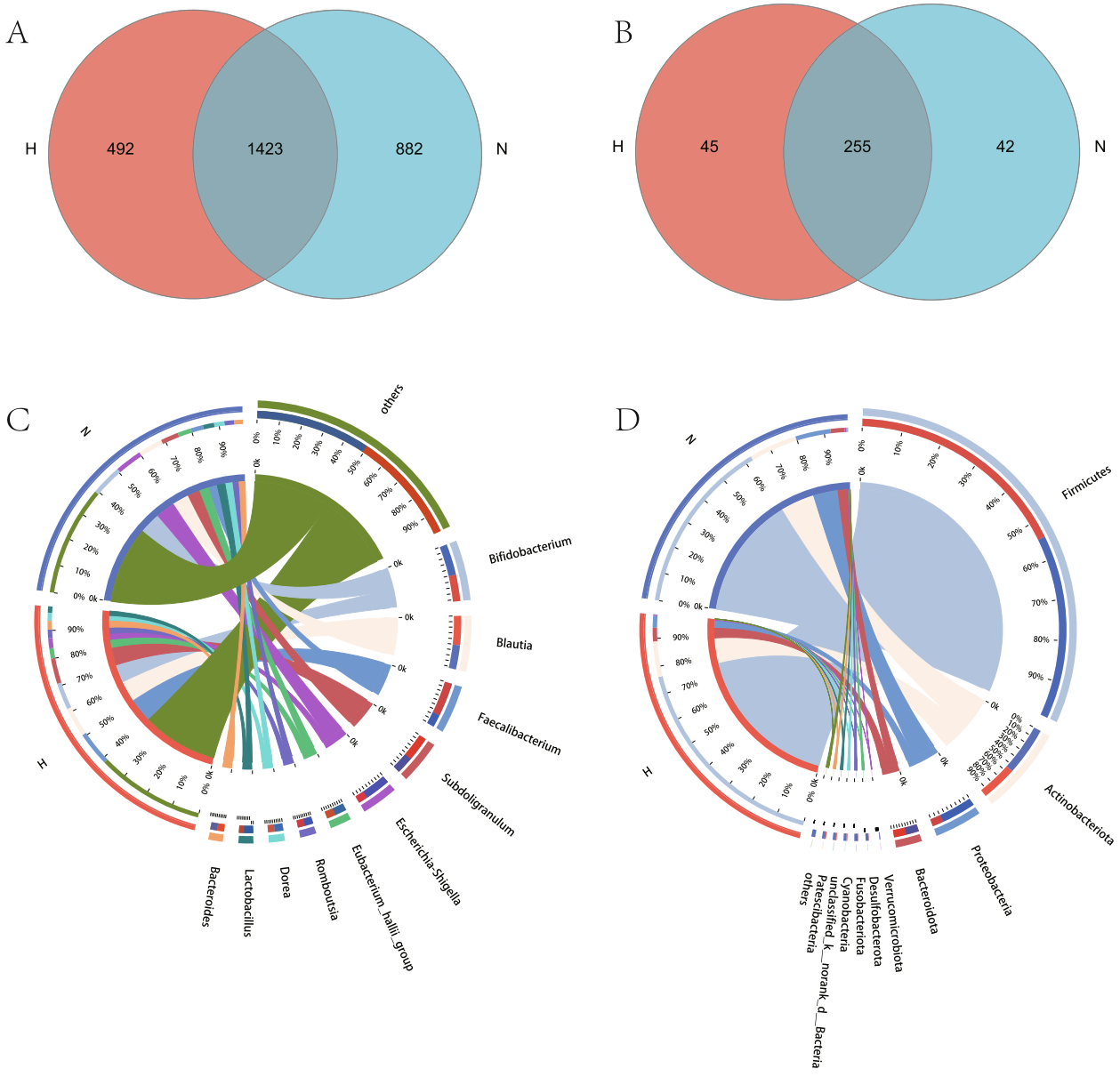

Fig. 2 Relationship between N and H gut microorganisms. **A** Common and unique ASVs for N and H. **B** The common and unique genus for N and H. **C** Comparison of N and H in the gut tract at the phylum level. **D** Comparison of N and H in the gut tract at the Genus level. These visualizations help reveal distinct differences in microbial community structure between the two groups across hierarchical taxonomic levels, providing insights into compositional shifts associated with cerebral infarction

microbiota did not exhibit significant differences between the two groups, the relative abundance of certain microbiotas at the phylum and genus levels was notably different in N compared to H, as shown in Fig. 2C and D.

In an effort to identify specific bacterial taxa associated with cerebral infarction patients, a linear discriminant analysis effect size (LEfSe) was utilized to compare the gut microbiota between the N and H groups. The analysis revealed significant differences in a total of 72 taxa. The H group exhibited a higher abundance of unique microbiota, while 35 gut microbiome taxa, including Actinobacteriota, Peptostreptococcales-Tissierellales, Coriobacteriia, Coriobacteriales, Anaerovoracaceae, Microbacterium, and others, were found to be significantly enriched in the N group, as presented in Fig. 3A and B.

The minimum cross-validation error was achieved with the inclusion of 22 key classes. The number of classes that minimized the cross-validation error curve stabilized when using these 22 classes, as shown in Fig. 3C and D. These 22 genera were proposed as microbiome markers to differentiate patients with cerebral infarction from healthy controls. The accuracy of this distinction was demonstrated with areas under the receiver operating

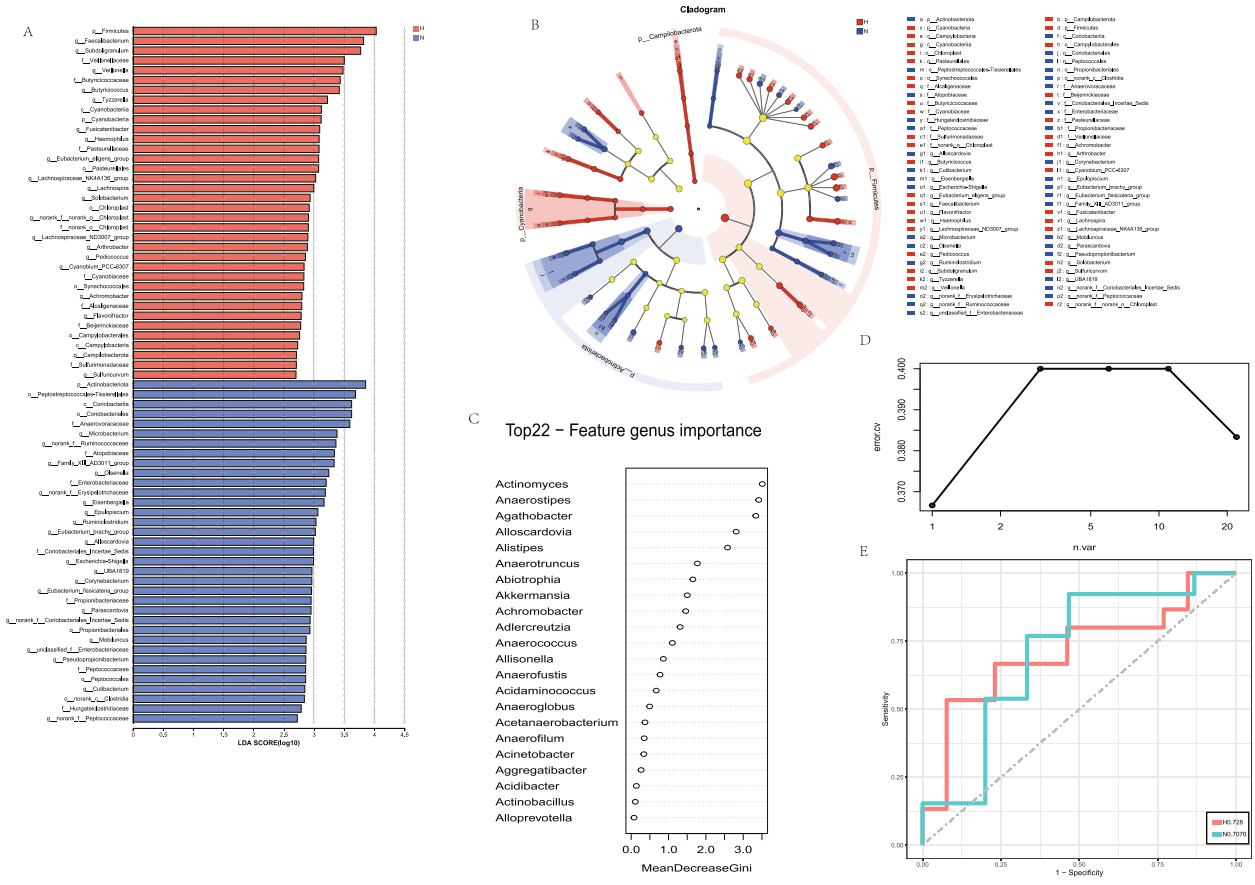

Fig. 3 Different genera as biomarkers in relative abundance identified. **A** The length of the column represents the influence of significantly different species in relative abundance (LDA scores > 4). **B** The significantly different species are shown in the cladogram. Each circle represents the phylogenetic level from phylum to genus inside to outside. Each circle's diameter is proportional to the taxon's abundance and the biomarker is consistent with the group marked with color. **C** The top 22 bacteria were identified by applying random-forest classification of the relative abundance of the gut microbiota in Cerebral infarction patients and Healthy. Biomarker taxa are ranked in descending order of importance to the accuracy of the model based on mean decrease Gini index; **D** 22 repeats of tenfold cross-validation error. **E** The predictive receiver operating characteristic (ROC) curves generated using the 22 candidate biomarkers contributing to Cerebral infarction. Area under the ROC Curve (AUC) (H, 0.728) > AUC (N, 0.7076)

characteristic (ROC) curves of 0.728 for the N group and 0.7076 for the H group, as depicted in Fig. 3E.

Subsequent profiling of the relative abundance identified 22 genera, one of which was found to be enriched in the N group. The outcomes of this profiling, as shown in Fig. 3, were consistent with the results of the microbial composition analysis presented in Fig. 2. Utilizing these 22 genera as microbiome markers effectively discriminated the N group from the H group.

Metabolomic analysis of N Vs H

To delve deeper into the pathogenesis underlying condition N, a nontargeted metabolomics approach was applied utilizing ultra-performance liquid chromatography coupled with quadrupole time-of-flight tandem mass spectrometry (UPLC-QTOF-MS/MS). The quality control (QC) samples exhibited a tight clustering pattern, which is indicative of the high stability and reliability of the metabolomics analytical system. Additionally, a discernible distinction between the N and H groups was observed in the analysis, signifying their metabolic divergence (Fig. 4A).

Employing Orthogonal Projections to Latent Structure Discriminant Analysis (OPLS-DA), a supervised pattern recognition technique, we were able to visualize and outline the overarching metabolic discrepancies between the N and H groups (Fig. 4B). This method aids in the identification of metabolites that contribute significantly to the group separation. Potentially differential metabolites were selected based on the Variable Importance for the Projection (VIP) score derived from the aforementioned OPLS-DA models.

A total of 30 endogenous serum metabolites were robustly identified and proposed as candidate biomarkers, distinguished by their VIP scores greater than 1.0, along with meeting the statistical significance threshold of $p < 0.05$ and undergoing mass spectrometry verification (Fig. 4C). These metabolites, highlighted in red in

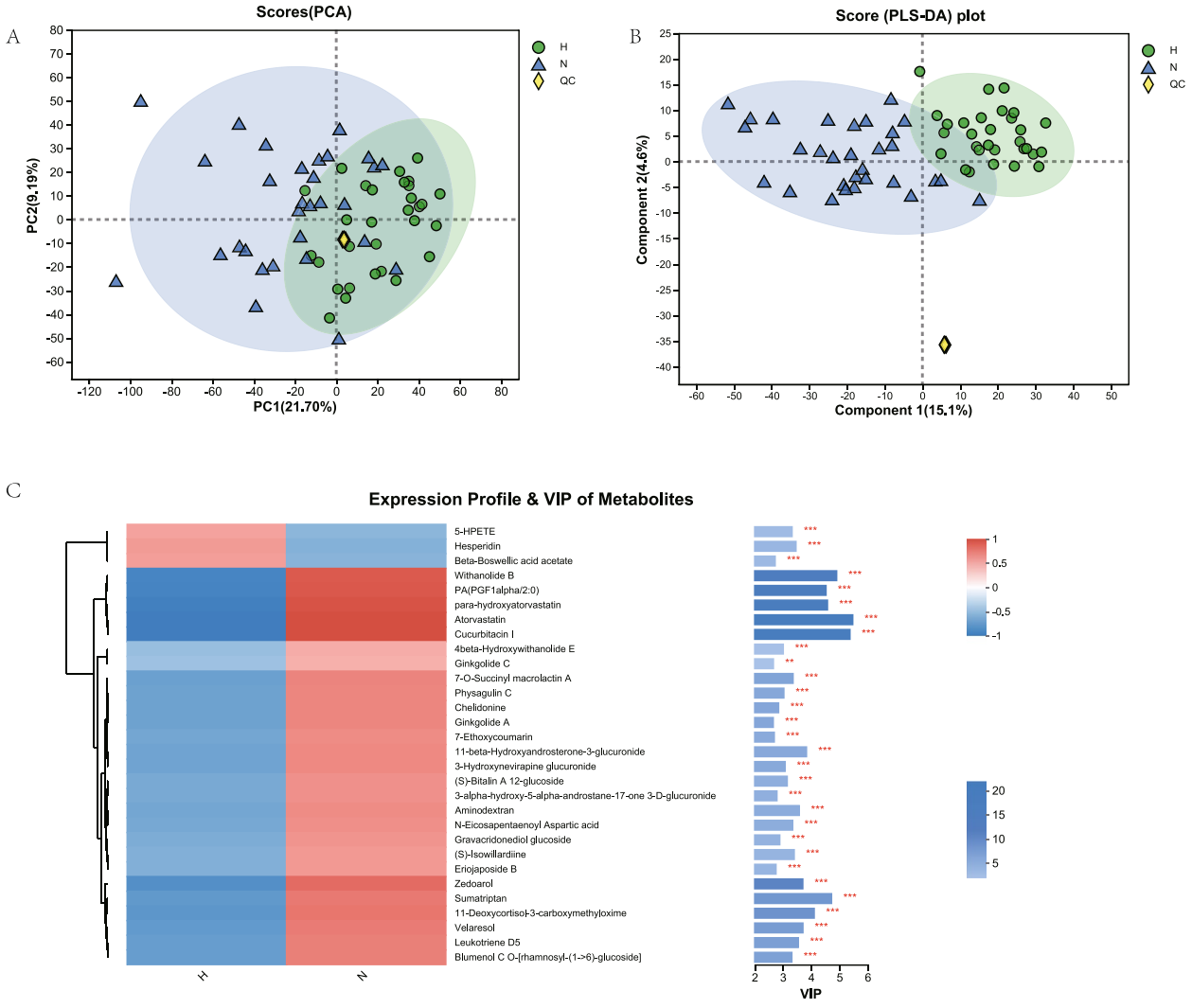

Fig. 4 Serum metabolomic profile in different groups. **A** PCA score plots of serum metabolic profiling of N and H groups. **B** PLS-DA score plots of serum metabolic profiling of N and H groups. **C** The relationship of Spearman between various groups and their metabolites. A red hue indicates a positive relationship, whereas a blue one suggests a negative one. The vibrancy of the color correlates with the intensity of the Spearman correlation. * $p \leq 0.05$, *** $p \leq 0.001$

the S-plots, represent the most influential factors in the metabolic variance observed between the groups.

Microbiome, metabolome and clinical indicators association analysis in N

In our investigation of the interplay between the gut microbiome, metabolome, and host physiology, we conducted a Spearman correlation analysis to assess the relationships between the 22 distinctive bacterial genera, 5 host clinical indicators, and 25 metabolites (Fig. 5). Our findings revealed positive correlations between the H group-enriched bacterial genera, their associated clinical markers, and VIP metabolites, suggesting congruence between gut microbial metabolic activity and the host's systemic metabolic and immune profile.

Furthermore, within the N group, robust positive correlations were observed between specific microbial genera, key clinical indicators , and serum metabolites, highlighting intricate tripartite interactions among the gut microbiota, host physiological status, and systemic metabolism.

Discussion

Recent research indicates that the intestinal microbiota and the nervous system interact through the brain-gut axis, exerting regulatory effects on neurological diseases [27, 28]. Berberine has been studied for its potential to modulate intestinal homeostasis and to treat ischemic stroke by influencing the gut microbiota and promoting the production of butyrate, a key metabolite [29]. Findings suggest that berberine preconditioning and

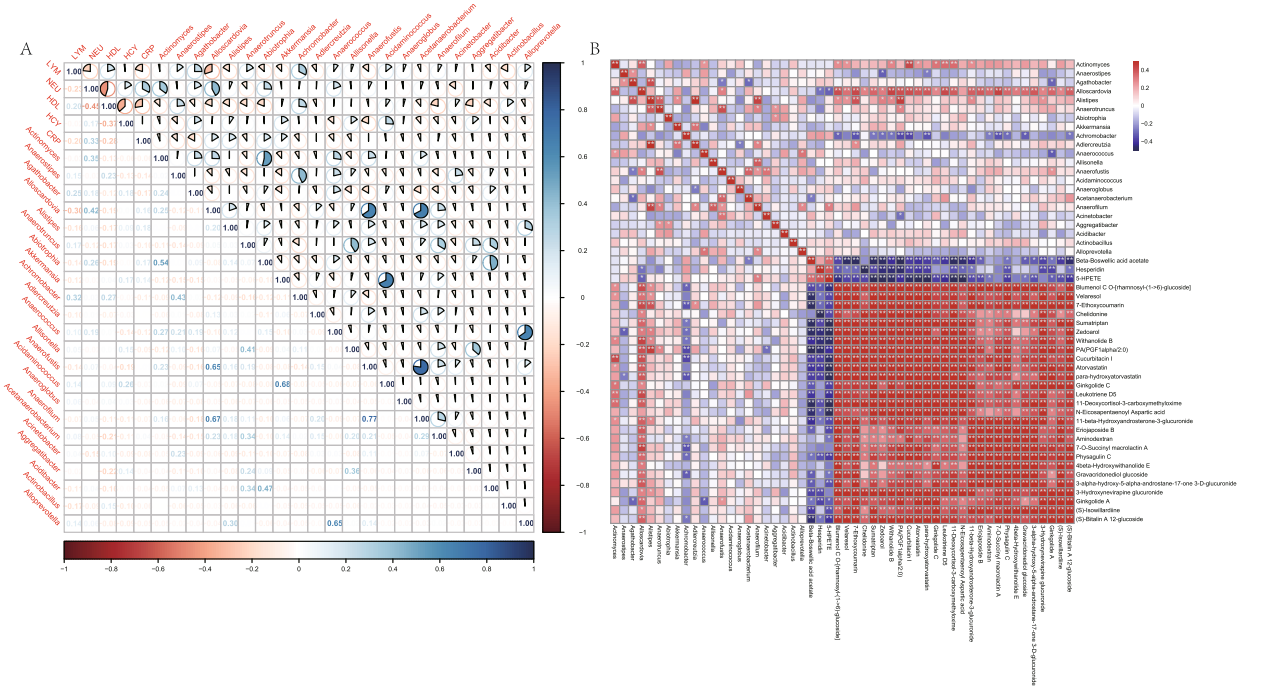

Fig. 5 The relationship between metabolite levels,clinical indicators and the gut microbiome. **A** Spearman correlations between 22 gut bacterial genera and 5 clinical indicators. **B** Spearman correlation analysis evaluated 22 bacterial genera and 30 metabolites. H-group-enriched bacteria correlated positively with its VIP metabolites. The N-group exhibited significant positive correlations between specific microbial genera and clinical indicators, as well as between microbial genera and serum metabolites, highlighting complex interactions.

treatment can enhance neuronal structure and function, increase gut microbiota diversity, and decrease neuroinflammation, thus offering neuroprotection. Additionally, berberine inhibits glial cell activation and the expression of NLRP3, reinforcing its neuroprotective role.

The connection between intestinal microbiome characteristics and stroke risk has been examined through risk stratification methods. High-risk stroke participants showed an enrichment of opportunistic pathogens, a depletion of butyrate-producing bacteria, and reduced fecal butyrate levels. Our results directly align with this paradigm: significant β-diversity differences in gut microbiota structure were observed between cerebral infarction patients and healthy controls, indicating that divergent community compositions likely drive distinct metabolic and immune responses. Specifically, 72 microbial taxa were identified as significantly different, with 35 taxa—including Actinobacteriota and Peptostreptococcales-Tissierellales—enriched in healthy controls, which may underpin the reduced short-chain fatty acid production and heightened pro-inflammatory states observed in patients. These findings align with prior observations that healthy individuals exhibit higher gut microbial diversity, which supports metabolic function and immune balance through mechanisms such as enhanced SCFA production. Conversely, cerebral infarction patients displayed reduced microbial diversity, with significant increases in specific taxa that may contribute to harmful metabolite accumulation and disease progression. In diabetic patients, an increase in lactobacillus and bifidobacterial levels and a decrease in butyric acid-producing genes were observed, which may influence stroke risk but requires further confirmation [3, 30, 31]. In 2023, Lou et al. reported no significant difference in microbial structure between acute ischemic stroke patients and healthy controls; however, significant differences were found in beta-diversity analysis, particularly when comparing patients with high platelet reactivity (HTPR) to those without (NHTPR), revealing distinct microbial genera between the two groups [32].

Using a random forest model, we identified 22 key microbial taxa as potential biomarkers for differentiating cerebral infarction patients from healthy controls, with diagnostic efficacy validated via ROC curve analysis. This reinforces the utility of gut microbiota profiling in clinical discrimination, suggesting that microbial community shifts could serve as noninvasive indicators of stroke risk. Lee et al. also discovered that estrogen may mitigate intestinal barrier damage, enhance intestinal microbiota composition, and decrease intestinal nervous activity post-stroke, potentially through the action of a microbiota-derived metabolite [33]. This suggests that microbial metabolites could be implicated in the pathogenesis of cerebral infarction. Our study identified differences in the intestinal microbiota of cerebral infarction patients compared to the normal population and a correlation with

blood metabolites. These findings suggest a significant role in the pathogenesis, progression, and prognosis of cerebral infarction, providing additional evidence for the brain-gut axis's influence on neurological diseases and offering potential diagnostic and therapeutic targets.

Metabolomic studies have identified plasma phosphatidylcholine metabolites, including choline, TMAO, and methionine, which are associated with cardiovascular disease. Our metabolomics analysis further revealed 30 serum metabolites as potential biomarkers, with significant intergroup differences. Notably, certain microbial genera in the cerebral infarction group exhibited strong positive correlations with specific serum metabolite concentrations—most notably TMAO—underscoring the intricate interplay between gut microbiota and systemic metabolism. TMAO, produced by gut microbes from dietary choline, promotes macrophage activation and atherosclerotic plaque instability, directly linking microbial dysbiosis to the pathological cascade of ischemic stroke: atherosclerosis, a well-established precursor to stroke, is exacerbated by TMAO-mediated endothelial dysfunction and thrombus formation, as demonstrated in preclinical models [10, 34]. Experimental evidence suggests that mice supplemented with choline or TMAO exhibit a significant increase in stroke volume, highlighting its role in elevating stroke risk [35, 36]. Gut microbiota can influence the development of arteriosclerosis through metabolites such as TMAO and phenylalanine, emphasizing the clinical relevance of this microbial-metabolite axis.

Importantly, our findings highlight the reduction of butyrate-producing bacteria in cerebral infarction patients, a trend linked to elevated stroke risk, as supported by prior studies [29, 30]. The diminished functional state of gut microbiota in patients may exacerbate neuroinflammation and impair recovery, emphasizing the need for interventions targeting microbial balance [37]. Post-ischemic stroke, changes in intestinal flora diversity and composition have been observed, with some harmful metabolites increasing [38]. Clinically, these findings suggest that microbiota-targeted interventions—such as probiotic therapies to restore butyrate-producing taxa or prebiotic diets to enhance SCFA production—could modulate neuroinflammation and improve stroke outcomes [39, 40]. For example, SCFAs like butyrate cross the blood–brain barrier to reduce activated microglia, suppress pro-inflammatory IL-17 + γδ T cell infiltration, and enhance synaptic plasticity, directly facilitating neural repair after stroke [13]. Recent research has demonstrated that upregulating short-chain fatty acids in the brain and increasing indolelactic acid further validate the therapeutic potential of microbially derived metabolites [14, 41], positioning the gut microbiota as a tractable target for improving post-stroke recovery. The gut microbiota's role in post-stroke recovery is thus pivotal, extending beyond metabolic support to actively modulate neuroinflammatory pathways critical for functional restoration.

This symbiotic relationship extends beyond mere metabolic interactions. Emerging research highlights a complex network of signaling pathways, indicating a profound interplay between the gut microbiota and metabolism. These insights pave the way for a deeper understanding of the molecular conversations that govern the subtle equilibrium between microbial communities and the host's physiological responses. As our exploration of this intricate interplay advances, we uncover the significant influence of these microbial orchestrators on the elaborate harmony of life's processes.

Advanced age is a significant risk factor for ischemic stroke, alongside other conditions such as hypertension and diabetes. However, the specific physiological dysfunctions that predispose the elderly to ischemic stroke are not fully understood. Some researchers have investigated the link between aging and ischemic stroke, noting that elderly mice exhibit diminished neurological function and heightened inflammatory responses following transient cerebral ischemia, leading to exacerbated physiological impairments [13]. Notably, the transplantation of aged mice's gut microbiota into young mice resulted in deteriorated neurological outcomes, poorer survival rates, and elevated inflammation. Crucially, post-ischemic stroke, the gut microbial community structure of elderly mice was observed to shift, intensifying their response to ischemia and increasing neurological damage [42, 43].

It has been reported that disruptions in the gut microbiota are associated with cognitive decline and a higher risk of post-stroke dementia. This may be mediated through mechanisms that trigger brain inflammation, compromise the blood–brain barrier, and accelerate neuronal aging [44–46]. In a clinical application, some domestic experts have used a novel Chinese medicine to treat ischemic stroke patients. The findings suggest that NaotaifangIII may offer neuroprotection against ischemic injury via the microbial-brain axis LPS/TLR4 signaling pathway, characterized by its ability to mitigate cerebral ischemic damage, reduce cerebral edema volume, and modulate gut microbiota balance, thereby inhibiting oxidative stress and inflammation [47]. Similarly, the decoction demonstrated efficacy in repairing the blood–brain barrier by upregulating tight junction proteins and reducing serum markers of brain injury [13, 38]. Flow cytometry analysis has shown that antibiotic-induced alterations in gut microbiota can upset immune homeostasis in the small intestine of mice. This change involves modulating dendritic cell activity, leading to an increase in regulatory T cells and a decrease in IL-17 + γδ

T cells, which in turn reduces neutrophil infiltration in the brain and decreases inflammation [48, 49]. Furthermore, research exploring the causal links between gut microbiota and cerebrovascular diseases has substantiated the impact of comorbidities such as obesity, alcohol consumption, and hypertension on stroke risk. Certain gut microbes have been found to have positive correlations with cerebrovascular diseases, while others have negative correlations [50]. Experiments with C57BL6/J mice have also indicated that gut microbiota imbalances are associated with an increased risk of Alzheimer's disease [45].

Despite these findings, the research has its limitations. The use of 16 s rDNA gene sequencing, while prevalent for identifying microbiota, may not fully capture the breadth of genetic characterization. Other factors, including diet and lifestyle, influence serum metabolite levels, making it difficult to attribute the origin of metabolites without isotopic dietary labeling. Additionally, the study's sample size was limited, and the data may not be representative, as it was collected from a single center. Therefore, further research incorporating metagenomics and metabolomics with larger, multi-centered samples is necessary to validate these results. Nonetheless, this study contributes valuable insights into the interplay among the gut microbiome, serum metabolome, and cerebrovascular diseases, guiding future research into intestinal biomarkers and targeted therapeutic strategies.

Abbreviations

IL-17	Interleukin 17
IFN-γ	Interferon-gamma
DNA	DeoxyriboNucleic Acid
16S rDNA	16S ribosomal DNA
PCR	Polymerase Chain Reaction
WBC	White blood cell
LYM	Lymphocyte
NEU	Neutrocyte
MONO	Monocyte
RBC	Red blood cell
PLT	Platelet
HGB	Hemoglobin
LDL	Low density lipoprotein
HDL	High density lipoprotein
UA	Uric acid
GLU	Glucose
HCY	Homocysteine
TG	Triglyceride
NLRP3	NOD-like receptor thermal protein domain associated protein 3
TMAO	Trimetlylamine oxide
LPS/TLR4	Lipopolysaccharide/Toll-like Receptor 4

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Authors' contributions
Wei Huang, Yinghui Chai, Xiang Li, Qiuyue Zhang, Zengkui Yan, Yan Wang, Xiaoyong Tao, Jiatang Zhang, and Feng Qiu all contributed to the research work. In this study, Wei Huang, Yinghui Chai, and Xiang Li made equal contributions to the research work. They had joint contributions in research design, data collection, analysis, or writing the paper. Xiaoyong Tao, Jiatang Zhang, and Feng Qiu are corresponding authors. They played key roles in guiding the research, obtaining funds, writing the paper, and submitting it. Other authors Qiuyue Zhang, Zengkui Yan, and Yan Wang participated in specific research work such as experimental design, data collection, and analysis.

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Data availability
The accession number of the SRA dataset in NCBI is PRJNA1089552. The accession number of the metabolome dataset in OMIX is PRJCA031294.

Declarations

Ethics approval and consent to participate
The protocol of this study was approved by the Ethical Committee of PLA General Hospital (HZKY-PJ-2020–38) and conducted in the principles and rules of the Declaration of Helsinki. All volunteers were provided with relevant information regarding their participation in the study and have signed written informed consent forms.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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